

HARSH SANWAL

Data Analyst | Machine Learning & Statistical Modeling Engineer

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Technical Skills

Programming Languages: Python, SQL, Node.js, Javascript

Analytics: Statistical Analysis, EDA, Forecasting, Predictive Analytics, RCA, Trend Analysis, Data Modeling, Data Validation, Data Quality, Data Governance

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, RAG, AI Agents, Prompt Engineering, ARIMA, Anomaly Detection, LLM Evaluation

BI & Visualization: Power BI, DAX, Power Query, Tableau, KPI Dashboards, Data Visualization

Databases & Engineering: PostgreSQL, Neo4j, ETL/ELT, Data Warehousing, FastAPI, REST APIs

Cloud & Tools: AWS, GCP, Docker, Git, CI/CD, ServiceNow, Salesforce, SharePoint, Microsoft 365

Work Experience

DRDO

Hyderabad, IND

** Incoming AI Research Intern*

Garvish Marketing Pvt. Ltd.

Dehradun, IND

Product & Data Analyst | SQL · Python · ServiceNow · Node.js · React

2024 – 2026

- Automated incident analysis using Python & SQL with ServiceNow API with 20% reduction in repeat incidents, 30% workflow automation.
- Built Python ETL pipeline + Power BI dashboard (5 KPIs: SLA, volume, resolution time, first-contact rate, MTTR) → 40% faster reporting.
- Collaborated with operations & UI/UX teams on process optimization.

Key Projects

FutureScope | Time-Series Forecasting & Scenario Analysis Platform *[Python · FastAPI · Redis · Kafka]*

- Architected Prediction, Scenario Generator, Simulation, Optimization engines platform for multi-domain future analysis
- Built FastAPI services generating 20 probabilistic scenarios (68%/95% CI); Neo4j knowledge graphs with explainability
- Validated PoC on urban water & traffic datasets; Agent simulation & Monte Carlo scaling; Second domain validation.

Supply Chain Disruption Detection & Root Cause Analysis *[Python · ETL · TensorFlow (LSTMs) · Anomaly Detection]*

- Unified 4 datasets into PostgreSQL schema with 20+ engineered KPIs, replacing silos.
- Built LSTM anomaly detection with correlation drift monitoring & statistical process control → 60% faster detection
- Delivered Power BI root cause dashboard with drill-down → 40% faster diagnosis for supplier escalation & rerouting.

Research

Defence & Multi-Agent AI Systems *(*Research In Progress)* | [Draft]

- Designing D-MAIDS (10-agent system); domain-specialist agents coordinate via structured prompting with human-in-the-loop review.

Education

BITS Pilani

M.Sc. (Master of Science), Data Science & AI

2026 – 2028

Panjab University

B.Sc. (Bachelor of Science), Physics

2023

Certifications

Anthropic Academy: Claude API, Claude Code, MCP, Prompt Engineering, AI Agents, Fluency, Safety & Evaluation

HarvardX: CS50's Introduction to CS

Udemy: Python Pro Bootcamp, Python & DSA, Generative AI (LangChain & Hugging Face)

Oracle: OCI AI Foundations Associate (2026), Oracle Agentic AI Foundations Associate (2026)

Achievements

Indian Army & IAF: Cleared the Officer Selection process on multiple opportunities, demonstrating leadership, teamwork, decision-making under pressure, and discipline.